Qassim University
College of Engineering
جامعة القصيم
كلية الهندسة

Introduction to Engineering Design-2
GE-213
Course First Day Material

(Active Learning Approach)

First Term 1448

(Fall 2026) Term 481

TABLE OF CONTENTS

		Page
1.	Course Philosophy and Content	3
2.	Course Structure	3
3.	Course Learning Objectives	4
4.	Classroom Expectations - The Need to Participate	5
5.	Professional and Ethical Behavior	6
6.	Assessment of Work and Course Grades	8
7.	Course Celebration	9
8.	Required Course Materials	9
9.	Motivation.....	11
10.	GE 213 Web site	11
11.	Course Portfolio	11
12.	Course Project	11
13.	Cheating in GE 213	11

1. Course Philosophy and Content

This course material is given to each student on the first day of class. The material contains useful information for students to enable them from understanding the philosophy of the course and acting effectively through the course.

GE 213 is a required introductory engineering course, which should be taken by all engineering students. The **COURSE GOAL** is to "***introduce students to the Engineering Methods.***" This is accomplished by focusing on:

i) Engineering Problem Solving, and
ii) Creating Computational Models.

In the same time the course represents a continuation of **GE 211 Course Goals** which include: Self-Regulation, Communication, Working cooperatively and collaboratively, and Quality. In addition, the following have been used to determine the course content:

1. The course must be of use to all engineering disciplines.

   We will focus on **the process of how to create computational models in order to solve problems**. We strive in this course to develop your confidence that you can develop computational models to solve problems, regardless of the nature of the problem.

2. The course must develop knowledge that is useful in sophomore level classes.

   GE 213 will prepare you to meet the general **presentation requirements** associated with technical work (i.e., what is expected when you turn in a homework assignment) and introduce you to **the process of creating purposeful models**. This will help you succeed with the discipline-specific problems to come later.

3. The course should help students evaluate whether engineering is a profession of interest.

   You can expect to:
   - use textbooks that present and discuss problems of the type engineers address,
   - receive general information about some specific engineering fields (e.g., what sort of problems a mechanical engineering graduate might confront), and
   - learn problem-solving approaches actually used in industry,

GE 213 course also addresses the following **OUTCOMES** of the study programs of Qassim University - College of Engineering:

1. an ability to identify, formulate, and solve engineering problems,
2. an ability to design a system, component, or process to meet desired needs,
3. an ability to use the techniques, skills, and modern engineering tools necessary for engineering practice,
4. an ability to communicate effectively,
5. an ability to function on multi-disciplinary teams, and
6. a recognition of the need for, and an ability to engage in life-long learning

2. Course Structure

The course content and classroom activities have been divided into two Modeling Sessions. You will attend each Session once per week. In this course you will spend 2 hours per week in Session_1 and another 2 hours per week in Session_2.

3. Course Learning Objectives

By the completion of this course the students should be able, when they face an unstructured (open ended) problem, to:

a. Apply Engineering Design Process:

1. ***Identify*** the most relevant needs (objectives) from an open ended problem **(Problem description – Comprehension).**
2. ***Discover*** a solution strategy using a suitable heuristic **(Road map planning – Application).**
3. ***Breakdown*** an open ended problem into its main elements (variables, constraints, parameters, etc.) **(Problem breakdown – Analysis).**
4. ***Develop*** alternative models by using basic mathematical, scientific and engineering knowledge **(Problem formulation – Synthesis).**
5. ***Choose*** the best model, from those developed, using specified criteria **(Model Assessment – Evaluation).**
6. ***Solve*** the model by a suitable computer software/tool and with an appropriate level of details **(Problem solving – Application).**
7. ***Test*** the model and ***analyze*** the results to determine if they are sufficient **(Solution validation – Analysis).**
8. ***Evaluate*** the solution and ***argue*** suitable improvements and changes **(Solution evaluation and improvement- Evaluation).**

b. Demonstrate Teamwork and Project Management:

9. ***Demonstrate*** work harmoniously and effectively in a team to solve open ended problems **(Team dynamics – Application).**
10. ***Demonstrate*** effective communication, ***resolve*** team conflicts, ***apply*** social team norms, ***prepare*** team rules, organize and delegate work as needed, and manage available resources **(Team Communication/Conflict Management –Application).**
11. ***Demonstrate*** time management for team meetings and to ensure that all tasks are completed and submitted on time **(Self-Management –Application).**

c. Display Communication Skills:

12. ***Write*** high quality design reports and ***prepare*** professionally the required graphs and tables. **(Written skills - Synthesis).**
13. ***Plan***, ***prepare*** and deliver clear and correct oral presentations using professional visual aids **(Oral skills- Synthesis).**

d. Develop Life-Long Learning Attributes:

14. ***Collect*** information of a new content by asking key questions and by using a variety of sources such as internet, textbooks, etc. **(Facing new concepts - Synthesis).**
15. ***Evaluate*** personal performance and progress as well as that of teammates using specific criteria **(self and peer evaluation – Evaluation).**

4. Classroom Expectations - The Need to Participate

The course uses Active Learning procedures, student teams, and the concept of continuous improvement of the learning process[1]. Constructivism[2] has also been used in developing a number of the in-class activities. Thus, ***you can expect to:***

- do a lot of talking while in class, both to small groups and to the entire class;
- be given problems that you think (at first) you do not know how to solve;
- find some of the classroom instructions to not be immediately clear;
- find that other students have different interpretations of what is expected during class;
- help other students understand course material at the request of the course faculty;
- be regularly invited to suggest ways to improve the class;
- be asked by a team member to modify the way you behave; and
- be quite active during most class periods.

It has been the course faculty's experience that students, who participate, even when they occasionally falter, have a much easier and more enjoyable time than students who resist participating. Your participation is a part of your course grade (see Section 6).
During each modeling session we expect that each student will:

1. Complete the assigned readings and activities before class.
2. Adhere to any announcements related to the course.
3. Attend and be on time for the scheduled class meetings.
4. Actively participate in all learning activities and tasks.
5. Treat his teammates with respect.
6. Avoid disruptive behaviors in class (e.g., side conversations, sleeping, etc.)
7. Meet or Exceed Expectations on the work products (i.e., Modeling Assignments).
8. Submit a Problem Solving and/or Reflective Writing Assignment to remove ONE defect (optional).
9. Improve work that is assessed as (NI), (NCE), or (NS).
10. Retain all of his work for the Modeling Sessions in a "Course Portfolio" until final grades are posted.

For each Modeling assignment there is an associated "Expectations" document and a "Checklist". The expectations document provides the details of what is required for each assignment and the checklist explains how the work will be evaluated. **These documents are provided in the Course web site and have to be printed before the session of the corresponding assignment.**

[1] Refer to the Blue Workbook which you studied in the course GE 211

[2] A simple constructivist class exercise is organized as follows: teams are assigned a problem but are not given any suggestions on how the problem might be solved. The teams work until they are unable to proceed. At that time the instructor provides either some guidance or teams share (with other teams) their views on how to proceed. The teams then continue working on the problem and the cycle is repeated.

5. Professional and Ethical Behavior

Professionalism is an approach and set of behaviors that all engineers are expected to exhibit as they pursue their profession. Some of these behaviors have been explicitly codified (e.g., Ethical Cannons of Engineering) but other, less explicitly stated expectations should still be observed (e.g., use of appropriate language). An important part of your education is to become aware of, and to begin practicing, these professional behaviors.

GE 213 includes four specific behavioral expectations: Academic Integrity, Appropriate Language, Appropriate Classroom Behavior and Appropriate Use of Qassim University Property and Facilities.

5.1 Academic Integrity

The University requires each student to act with honesty and integrity and to respect the rights of others in carrying out all academic assignments. There are a number of actions that constitute a violation of the policy. These actions include, but are not limited to:

- referring to unauthorized materials or courses or employing unauthorized devices (e.g., audio recorders, crib sheets, calculators, solution manuals, or commercial research services) during tests, quizzes, homework, and class activities;
- possession, buying, selling, or otherwise obtaining or using, without appropriate authorization, a copy of any materials intended to be used for academic evaluation in advance of its administration;
- depending on the aid of others to the extent that the work is not representative of the student's abilities, knowing or having good reason to believe that this aid is not authorized by the instructor;
- submitting the ideas or work of another person or persons without customary and proper acknowledgment of sources, and
- permitting one's own ideas or work to be submitted by another person without the instructor's authorization.

University policy allows for cheating sanctions ranging from zero credit for an assignment to expulsion (without expectation of readmission) from the University.

Any student who is found to have violated the University's Academic Integrity Policy in GE 213, no matter how minor the violation, will have the sanctions set by the Engineering College Policy on cheating. The sanction could result in expulsion with a grade of FX, failed because of cheating, entered on your transcript.

5.2 Appropriate Language

Engineers are expected to effectively communicate ideas. Inappropriate language (written and oral) does not effectively communicate your ideas to an audience. In **GE 213** you are expected to use ap- propriate language in:

- all your written communications (assignments),
- all your oral communications (presentations, conversations with peers or instructors), and
- all your electronic communications (e-mail).

Use of inappropriate language will result in the issuance of a Self Regulation Lapse; repeated use of inappropriate language will result in your withdrawal from the class.

5.3 Appropriate Classroom Behavior

In **GE 213**, you are expected to participate in the various classroom activities, including:

1. coming to each class on time,
2. working on whatever assignment has been given,
3. being awake,
4. not consuming any food or drink (even water) while in the Qassim University classrooms,
5. following instructions given by the instructor, and
6. avoiding disruptive side conversations.

Repeated occurrences of inappropriate classroom behavior will result in a reduction in your grade (see "Self Regulation Lapses" in Section 6.

As **GE 213** based on Active Learning approach, instructors use **classroom observations** to gauge how well the student groups are interacting, if they are on task, how well the in-class activity went, the quality of classroom discussions and interactions between students and faculty. The instructors may give extra marks (***Bonus***) to the students or groups who communicate effectively inside the class. Themaximum numbers of bonuses which can be given to any student are 5 per academic semester.

5.4 Appropriate Use of Qassim University Property and Facilities

In **GE 213**, you will use a variety of Qassim University facilities and handle a variety of Qassim University property. You are expected to make appropriate use of these facilities and property, including:

1. gently handling **all items** distributed for your use and returning such items in the same condition as they were issued: folders, artifacts, handout sheets, etc.;
2. leaving a clean work space – tables, floors and chairs;
3. treating furniture gently – avoiding rough treatment, putting feet on tables and chairs, etc., not writing upon or disfiguring furniture; and
4. leaving computers as you would furniture – clean and ready to use, without any remaining software, links, screen savers or settings that will offend or impede the efforts of subsequent users.

Especially serious abuse of Qassim University property and facilities can result in University-level sanctions, including prosecution. Repeated minor occurrences of inappropriate use of Qassim University property and facilities will result in a reduction in your grade (see "Self Regulation Lapses" in Section 6).

6. Assessment of Work and Course Grades

Most students find that assessment (grading) is the area in which they find the most discrepancy between their **expectations** of the course and the **reality** of the course. We wish to reassure students that the **GE 213** assessment process is carefully and rationally designed - it is organized to help stu- dents define and recognize **quality**, and to motivate students to achieve **quality work**. The follow- ing material will discuss how the course assesses your cognitive behavior, your affective behavior and how these two assessments are combined to generate a grade.

6.1 Assessment of Cognitive Behavior (What you Know)

Your grade for a work product is based on the **quality of work** you or your team submits. Quality work is defined as essentially complete and correct work that is documented neatly in a well-organizedfashion and submitted on time. To help you submit quality work, in each Session you will be referred to a set of Checklists that define what is expected for your assignments.

Submitted work is assessed either as Exceeding Expectations (**E**), Meeting Expectations (**M**), Needing Improvement (**NI**), No Credible Effort (**NCE**) or No Submittal (**NS**).

- To **meet expectations** (**M**) on any given assignment, you must meet the required expectations addressed on the assignment's checklist. Failure to meet these expectations wholly or partially results in a grade of (**NCE**) or (**NI**).
- To **exceed expectations** (E), the work must first meet expectations and then in addition, exhibit some relevant work beyond what was requested in the assignment. Elements of exceeding expectations might include demonstrating *a level of understanding* beyond that asked for in the assignment description, answering a *relevant question* not originally asked for, or doing some extraordinary level of achievement.
- Work that does not demonstrate at least a credible effort towards completion is assessed as no credible effort (**NCE**) and a *Self Regulation* ***Lapse*** is also recorded. When work is not submitted, an assessment of No Submission (**NS**) and a *Self Regulation* ***Lapse*** is also recorded.
- If work is assessed as **NI, NCE or NS**, you are given one time chance to make improvements and resubmit the work. Provided a credible effort is made to improve the work.
- After successful resubmission, work will receive, as a maximum, meet expectation. Otherwise, it will receive the same assessment if no credible effort had been made to improve the work.
- Resubmitted work must be accompanied by the original work.

All Modeling Work Products must follow the following format guidelines:

- Clear, concise, complete, accurate, organized work are key elements
- They must be computer generated (Excel, Word...) unless otherwise specified
- They must include your name and team number on each page
- They must be spell checked and demonstrate your best grammatical skills;
- Your model should be constructed using the Concept of “Sandwich” format.
- They must include a printed copy of the formula sheet
- They must be stapled together with the appropriate checklist completely filled out (i.e., the work is self assessed) and attached to the front. The checklist may be filled out by hand but must be completed.

6.2 Assessment of Affective Behavior (Self Regulation)

Independent of work quality, affective *behavior* (as described in Section 4) also impacts your **GE 213** grade. One of the course goals is that students demonstrate ***self regulation***[3]. Instances of poor behavior or participation are recorded as *Self Regulation* ***Lapses***. *Self Regulation* ***Lapses*** include, but are not limited to:

- absences from class,
- lateness to class up to10 min results in 1/2 laps and 1 laps after 10 min.
- failure to participate in class activities and/or being unprepared for class,
- disruptive or unethical classroom behaviors,
- failure to cooperate with in-class instructions,
- minor inappropriate use of Qassim University properties and facilities,
- getting no credible effort (NCE) or No submittal in the first submission for an assignment.
- Late submission is not permitted for first submission, in this case the work can be submitted in the resubmission due date
- late submittal of resubmitted work (1 laps for first week late, 2 lapses for the second week),

6.3 Course Grades - The General Philosophy

Course grades are based on the **final** quality of the assignments you submit[4]. If by the end of the semester you have not improved any NI's, NCE's, and/or NS's you received during the semester or if you have accumulated more than four Self-regulation lapses you will accumulate **Defects** which havea negative impact on your grade[5]. How the final assessed quality E, M, NI, NCE, NS and Defects are converted to the course grade, these can be found in the Tables 1 and 2 below. The tables demon- strates several important aspects of determining your course grades.

7. Course Celebration

There is **a Final Individual in Class Assignment** in **GE 213**; and it is called **The Celebration**. The time for the Celebration is shown on your Course Calendar. Celebration (Oral presentation) is considered as an assignment and is marked using the same scale of (E, M, NI, NCE, NS).

8. Required Course Materials and Accessories

The following is a list of items that you (or your team) are required to purchase.

1. *For each student. The course text book entitled: "How To Model It Problem Solving For The Computer Age," by* Anthony M. Starfield, Karl A. Smith, and Andrew L. Bleloch
2. For each team (wait to purchase until your team is formed):
   a) A course portfolio and dividers.
   b) Removable computer storage media (flash memory and a CD) and Stationary kit

[3] See Section B of the Blue Workbook for more information on Self Regulation.

[4] Work that is initially assessed as NI or NS but is assessed as M on resubmission is treated as M in determining grades.

[5] Because students sometimes accumulate too many Self-regulation lapses before they fully understand the consequences, students are provided an opportunity to remove **ONE** Defect in **GE 213**. If you do accumulate a **Defect** because of too many lapses in a Session and want to remove the **Defect**, you will need to see your Instructor for details on the assignment.

Table 1: Calculation of the Course Defects

Note: The 'Defects' are defined below; the impact of the 'defects' on your semester grade represents your Customer's sense of the relative importance of you demonstrating the characteristics of a Self-Regulated Learner. The 'defects' are cumulative.

PART I: Cognitive Problems

1. Final assessment of No Submittal "NS" or No Credible Effort "NCE"

No. of NCE or NS	1	2	3	4
No. of corresponding Defects	1	2	3	5

2. Final assessment of Needs Improvement "NI"

No. of Needs Improvement "NI"	1	2	3	4	5	6
No. of corresponding Defects	0.5	1	1.5	2	3	5

3. **Plagiarism Report**

Percentage of similarity >	35%	50%	60%	70%	80%	90%
No. of corresponding Defects	0.5	1	1.5	2	3	5

PART II: Affective Behavior Problems (self regulation issues)

No. of Self Regulation Lapses †	0 - 2.5	3-4	4.5-5.5	6 - 7	7.5-8	8.5-9.5	More than 9.5
No. of corresponding Defects	0	0.5	1	2	3	4	5

†Self Regulation Lapses are addressed in item 6.2 in this First Day material.

Table 2: Calculation of the Course "GE 213" Grades

		No. of Defects							
		0	0.5	**1**	1.5	**2**	**2.5**	**3**	**> 3**
No. of Exceeds	**≥ 3**	A^+	A	B^+	B	C^+	C	D^+	F
	2	A	B^+	B	C^+	C	D^+	D	
	1	B^+	B	C^+	C	D^+	D	F	
	0	B	C^+	C	D^+	D			

9. Motivation

Students who present distinguished performance are given motivation bonuses according to the opinion of the instructor. These bonuses positively affect the student grade as follows:

1. Each bonus removes one **self regulation lapse**
2. If the student has no self regulation lapses, **four or more bonuses** upgrade one assignment either **from NI to M or from M to E**.

10. GE 213 Web site

All documents used in GE 213 are available for downloading from the following web site:

https://lms.qu.edu.sa/webapps/login/

Most of the documents are MS Office files but several of the documents have .**pdf** extensions which mean they require the Adobe Acrobat Reader to view and print. On campus this will not be a problem but you may need to download a free copy of the Acrobat Reader for your home computer.

You will need to print all the checklists, assignments and expectations that are used in the course but not necessarily much more. Before you print anything ask yourself:

"Why do I need a hard copy of this document?"

"Will I likely keep this material past the end of this week, the end of this semester?"

11. Course Portfolio

You are expected to retain all of your work for the Modeling Sessions (including class work and presentations) in a "Course Portfolio". The course portfolio is marked as an assignment and has an Expectations document and a Checklist. You will be asked to submit your team portfolio several times during the semester before the last submission indicated in Calendar.

12. Course Project

You will only learn the tools and techniques of engineering design if you apply them yourself. Therefore, you will be expected to apply the techniques and methods covered in the course to a modeling project. The deliverables of this project are two: a ***formal public presentation***, and a ***technical report*** to the client that documents both the process each team followed and the design outcome that each team proposes. Each member of the team will be expected to have made ***significant contributions to the project***; individual effort and contribution will be evaluated both by the project's faculty advisor and fellow team members (peers).

13. Cheating in GE 213

Some of the assignments in the **GE 213** Modeling section evaluate your individual work. However, during class, you will be generating ideas with your teammates and classmates. Therefore, all of the ideas generated by your team in class may be reproduced and included in your individual documentation. All of the information and ideas presented by other teams during class may be included in your individual documentation with appropriate attribution or reference. This material may not be reproduced or copied; it must be put in your own words.

You may meet with your teammates or classmates outside of class to work on your assignments. **However**, the final work product that you submit must be your individual effort and documentation (i.e., your final documentation, spreadsheets, and charts must be unique and may not exactly duplicate any portion of anyone else's work).
Not following the above instructions is considered as cheating and the university cheating policy will be applied.